so $G^\vee \cong G$ for any finite abelian group G. This isomorphism is, however, noncanonical: the particular decomposition of G as $\mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}$ is not unique as far as which elements of G correspond to $\mathbb{Z}_{n_1}$, etc., is concerned. On the other hand, $G \cong (G^\vee)^\vee$ is a canonical isomorphism, given by $\varphi : G \to (G^\vee)^\vee$, where $\varphi(g)(\chi) = \chi(g)$.

(2) The symmetric group S_3. In the symmetric group S_n, conjugacy classes are determined by cycle decomposition sizes: two permutations are conjugate if and only if they have the same number of cycles of each length. For S_3, there are three conjugacy classes, so there are three different irreducible representations over $\mathbb{C}$. If their dimensions are d_1, d_2, d_3, then $d_1^2 + d_2^2 + d_3^2 = 6$, so S_3 must have two 1-dimensional and one 2-dimensional representations. The 1-dimensional representations are the trivial representation $\mathbb{C}_+$ given by $\rho(\sigma) = 1$ and the sign representation $\mathbb{C}_-$ given by $\rho(\sigma) = (-1)^\sigma$.

The 2-dimensional representation can be visualized as representing the symmetries of the equilateral triangle with vertices 1, 2, 3 at the points $(\cos 120°, \sin 120°)$, $(\cos 240°, \sin 240°)$, $(1, 0)$ of the coordinate plane, respectively. Thus, for example,

$$\rho((12)) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \rho((123)) = \begin{pmatrix} \cos 120° & -\sin 120° \\ \sin 120° & \cos 120° \end{pmatrix}.$$

To show that this representation is irreducible, consider any subrepresentation V. The space V must be the span of a subset of the eigenvectors of $\rho((12))$, which are the nonzero multiples of $(1, 0)$ and $(0, 1)$. Also, V must be the span of a subset of the eigenvectors of $\rho((123))$, which are different vectors. Thus, V must be either $\mathbb{C}^2$ or 0.

(3) The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, with defining relations

$$i = jk = -kj, \quad j = ki = -ik, \quad k = ij = -ji, \quad -1 = i^2 = j^2 = k^2.$$

The five conjugacy classes are $\{1\}, \{-1\}, \{\pm i\}, \{\pm j\}, \{\pm k\}$, so there are five different irreducible representations, the sum of the squares of whose dimensions is 8, so their dimensions must be 1, 1, 1, 1, and 2.